Question			Answer	Marks	AO	Guidance
9	(a)		$(\mathrm{f}'(x) =)\ \ 3x^2 + 3\mathrm{e}^{3x}$	B1 B1 [2]	1.1 1.1	• $3x^2$ or $k\mathrm{e}^{3x}$ where $k = 1, 3$ or $\frac{1}{3}$ • Implied by correct answer • Fully correct derivative
9	(b)		$(0, 1)$	B1 [1]	1.1	• Or $x = 0$, $y = 1$. • 1 alone is **not** sufficient. • Condone omission of brackets
9	(c)		$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\frac{\mathrm{d}x}{\mathrm{d}y}}$ so gradient is $\frac{1}{\textit{their}\text{ gradient of } y=\mathrm{f}(x)(\text{where it crosses } y\text{-axis})}$	M1	3.1a	• Consideration of relationship between gradients of the two curves where they cross the relevant axes. May be shown in a diagram with sketch of any increasing function reflected in $y = x$ • Implied by correct answer • **FT** *their* derivative
			Alternative method for M1 $x = y^3 + \mathrm{e}^{3y}$ $1 = 3y^2\frac{\mathrm{d}y}{\mathrm{d}x} + 3\mathrm{e}^{3y}\frac{\mathrm{d}y}{\mathrm{d}x}$ $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{3y^2+3\mathrm{e}^{3y}}$	M1		• Clear attempt to use implicit differentiation to find $\frac{\mathrm{d}y}{\mathrm{d}x}$ • Implied by correct answer
			$\frac{1}{3}$	A1 [2]	1.1	• nfww